Software Engineer - Site Controller, Energy Storage

Redwood Materials · San Francisco, California, United States

Job ID (this archive)	ENERGY-090
Employer	Redwood Materials
Location(s)	San Francisco, California, United States
Career stage (archive tag)	Entry or experienced (Entry / Mid-career)
Location group	Other U.S.
Majors (archive tags)	Electrical Engineering, Computer Science
Energy area	Battery storage / software
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/redwoodmaterials/jobs/6097367004?gh_jid=6097367004
Original link (from the course sheet)	https://boards.greenhouse.io/redwoodmaterials/jobs/6097367004?gh_jid=6097367004
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-061

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Redwood Materials

Redwood Materials was founded in 2017 by JB Straubel, co-founder and former CTO of Tesla, on a simple but radical idea: the critical minerals powering the energy transition already exist. They just need to come back. Today Redwood is the largest lithium-ion battery recycler in North America — recovering, refining, and redeploying critical minerals from end-of-life batteries back into the supply chain. On the energy side, Redwood has deployed the largest second-life battery project in the world, delivering power to AI data centers in months rather than years using proprietary energy architecture engineered entirely in-house. The company is profitable, the market it operates in is projected to grow 20 to 50 times over the next decade, and Redwood is the only company with the supply chain, the technology, and the operational scale to lead it.

Software Engineer – Site Controller, Energy Storage

We are at the precipice of a global energy transition. At Redwood Materials, we are moving beyond battery recycling to build the most advanced, circular energy ecosystem on the planet. Our goal is to move the industry away from static, hardware-heavy infrastructure toward agile, software-controlled energy assets.

The Site Controller is the "Central Nervous System" of our modular Battery Energy Storage Systems (BESS). It is the platform that allows us to harmonize new and repurposed batteries into a single, high-performance energy asset. We are looking for engineers to build the digital backbone that finally makes renewable energy baseload-ready and turns the grid into a truly digital, responsive network. As a Site Controller Software Engineer, you will be responsible for the high-level orchestration of our grid-scale storage sites. Operating at the "System Coordination" layer, you will design and implement the mission-critical logic that dictates how GWh-scale sites interact with the utility, industrial assets, and the Redwood Cloud. You will build the scalable, resilient software platform that makes our proprietary "Pack Manager" technology possible.

Responsibilities will include:

- Distributed System Architecture: Design and implement a scalable, resilient software architecture for the Site Controller, ensuring it can coordinate thousands of distributed microcontrollers with zero downtime.

- Tiered Control Logic: Lead the development of multi-rate control systems, balancing high-level site coordination with real-time grid execution. You will collaborate with simulation and controls teams to optimize state-of-charge and battery longevity at the site coordination layer, while leading the integration of high-speed closed-loop controls to ensure sub-cycle grid synchronization and precise command execution.
- System Resilience & Fault Handling: Architect robust fault-management frameworks, designing the state machines responsible for autonomous error recovery and failsafe modes.
- Deterministic Networking: Architect the physical and real-time networking layers. You will define the communication interfaces and network protocols that ensure low-latency, reliable data flow between the site controller, local assets, and upstream telemetry.
- Hardware Abstraction & Integration: Design and define the critical software interfaces that enable our "Pack Manager" to serve as a universal translator. You will architect the communication layers to ensure high-availability and resilient data exchange between the centralized site controller and a heterogeneous fleet of downstream microcontrollers.
- Testable Software Excellence: Design and implement highly modular, testable software. You are responsible for ensuring that all site controller logic is built with automated unit and integration testing in mind, providing a robust foundation for system-wide validation.
- System Diagnostics: Familiarity with distributed logging tools and optimizing telemetry data flows for real-time system health and performance tracking.

Desired Qualifications:

- Background: A BS or MS in Computer Science or Electrical Engineering, combined with 2+ years of professional experience developing application-layer software for hardware-constrained systems.
- Modern Toolchain: Proficiency in Rust and Python. You understand when to leverage Rust’s safety and performance for site orchestration and how to utilize Python for rapid tooling and simulation.
- Industrial Networking: Deep familiarity with industrial communication protocols (Modbus TCP and CAN). Experience with DNP3 or IEC 61850 is a significant plus.
- System Administration: Strong experience with Linux system administration and modern containerization technologies (Docker, Kubernetes) for edge computing.
- Fleet Orchestration: Experience implementing and managing high-availability Firmware Over-the-Air (OTA) pipelines, ensuring secure and reliable updates from the cloud through to downstream controllers.
- Data Lifecycle: Experience with time-series databases and data visualization tools for monitoring system health and energy flows.
- Hardware Integration: Proven track record of integrating vendor-off-the-shelf (VOTS) components—such as PV systems, industrial inverters, fire safety systems, and HVAC controllers—into a unified software platform.

In accordance with California pay transparency laws, the salary range for this position is listed below. Actual compensation may vary based on a variety of factors, including experience, education, and skills.

California Pay Range:

$180,000—$237,500 USD

The position is full-time. Compensation will be commensurate with experience.

We collect personal information (PI) from you in connection with your application for employment with Redwood Materials, including the following categories of PI: identifiers, personal records, professional or employment information, and inferences drawn from your PI. We collect your PI for our purposes, including performing services and operations related to your potential employment. If you have additional privacy-related questions, please contact us at privacy@redwoodmaterials.com.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Background: A BS or MS in Computer Science or Electrical Engineering, combined with 2+ years of professional experience developing application-layer software for hardware-constrained systems.

Key skills listed in the archive: Python; Rust; Linux; Docker; Kubernetes; time-series

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 6097367004

Career stage: Entry / Mid-career

Major fit: Direct skills pathway: software development, computing and engineering; consult the exact degree and experience requirements. Energy-sector connection: Control, integration and infrastructure software for stationary energy storage.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 6097367004.

Sources and retrieval details

Employer posting

https://boards.greenhouse.io/redwoodmaterials/jobs/6097367004?gh_jid=6097367004

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https://boards-api.greenhouse.io/v1/boards/redwoodmaterials/jobs?content=true

Retrieved: 2026-10-26T16:33:09.192514+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.